

Part E: Feature Engineering

Mixed variables

```
customer_id      int64
age              int64
gender           object
region           object
education_level   object
employment_type   object
annual_income     float64
loan_amount       float64
loan_purpose        object
credit_score      float64
repayment_history int64
transaction_count int64
spending_ratio    float64
join_date         object
default_flag      int64
income_missing    int64
dtype: object
```

Data & time handling

```
join_date  Year  Month  Day  Weekday
0 2020-01-01  2020     1    1  Wednesday
1 2020-01-02  2020     1    2  Thursday
2 2020-01-03  2020     1    3   Friday
3 2020-01-04  2020     1    4  Saturday
4 2020-01-05  2020     1    5   Sunday
```

Ordinal encoding

```
education_level  education_level_encoded
0      High School                1.0
1    Post-Graduate                2.0
2      Graduate                  0.0
3      Graduate                  0.0
4      High School                1.0
```

Label encoding

```
gender  gender_encoded
0  Female                0
1   Male                 1
2   Male                 1
3  Female                0
4   Male                 1
```

One hot encoding

```
pr > ml %>% gather(region) %>%
  region_East region_North region_South region_West
0      True      False      False      False
1     False      False      False      True
2      True      False      False      False
3     False      False      False      True
4     False      False       True      False
```

Binning

	annual_income	income_bin
0	731147.0	Medium
1	1301267.0	High
2	758988.0	Medium
3	296393.0	Low
4	1431266.0	High

Binarization quantile

	annual_income	income_binary
0	731147.0	1.0
1	1301267.0	1.0
2	758988.0	1.0
3	296393.0	0.0
4	1431266.0	1.0

K – means

	annual_income	income_cluster
0	731147.0	2
1	1301267.0	0
2	758988.0	2
3	296393.0	1
4	1431266.0	0